Deep Ordinal Classification

From nominal losses to unimodal output layers

Carla Monteiro, Valentina Corbetta · MICCAI 2026 ·

01.10.2026



A taxonomy of ordinal classification methods

Methods differ in how strongly they constrain the posterior distribution[1]

Nominal

Ignores the class order

CE

Non-unimodal

Uses the order; output shape unconstrained

OE

Soft-unimodal

Promotes unimodality through the training objective

Soft targets: BCE, Lp

CE + regulariser: Wu-Wass, MVL

Moment-based loss: exp_MSE

Hard-unimodal

Guarantees unimodality through the output layer

UN



Increasing constraint on the shape of the output distribution

[1] Cardoso et al. "Unimodal distributions for ordinal regression." IEEE Transactions on Artificial Intelligence (2025).

Notation

K ordered classes, indexed i = 0, …, K − 1; true class index k

The network maps an input x to logits v; the softmax gives the posterior

$$\hat{\mathbf{y}} = \mathrm{softmax}(\mathbf{v}) = [\hat{y}_0, \ldots, \hat{y}_{K-1}], \qquad \hat{k} = \arg\max_i \hat{y}_i$$

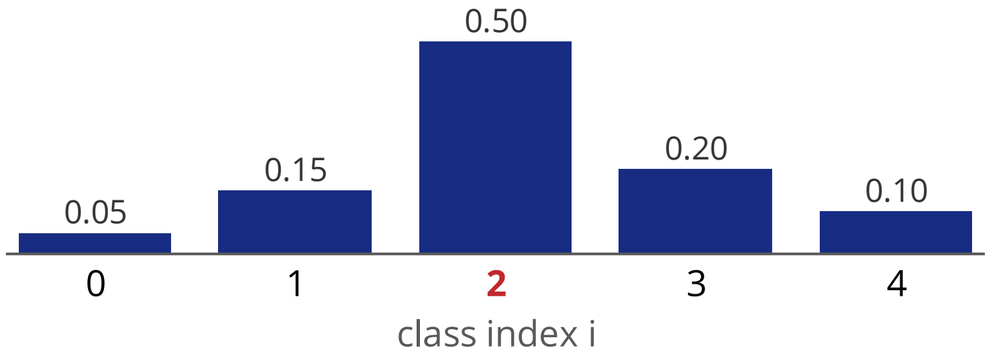
Running example (used throughout)

K = 5 classes (BI-RADS 1–5)

True class k = 2 (BI-RADS 3)

Class indices start at 0, as in the formulas

ŷ = [0.05, 0.15, 0.50, 0.20, 0.10]



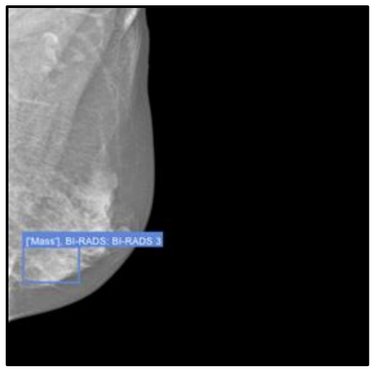
Baseline model

The standard loss function for multi-class categorical classification is the Cross-Entropy (CE) loss[2]

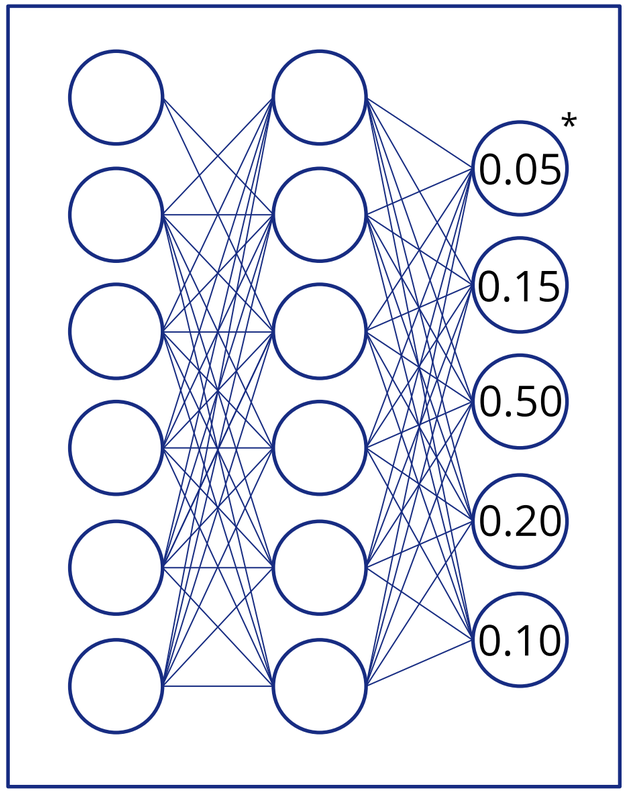
[2] Zhang et al.. "Generalized cross entropy loss for training deep neural networks with noisy labels." *Advances in neural information processing systems* 31 (2018).

Baseline model

The standard loss function for multi-class categorical classification is the Cross-Entropy (CE) loss[2]



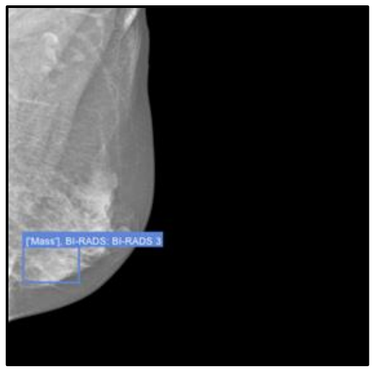
BI-RADS 3



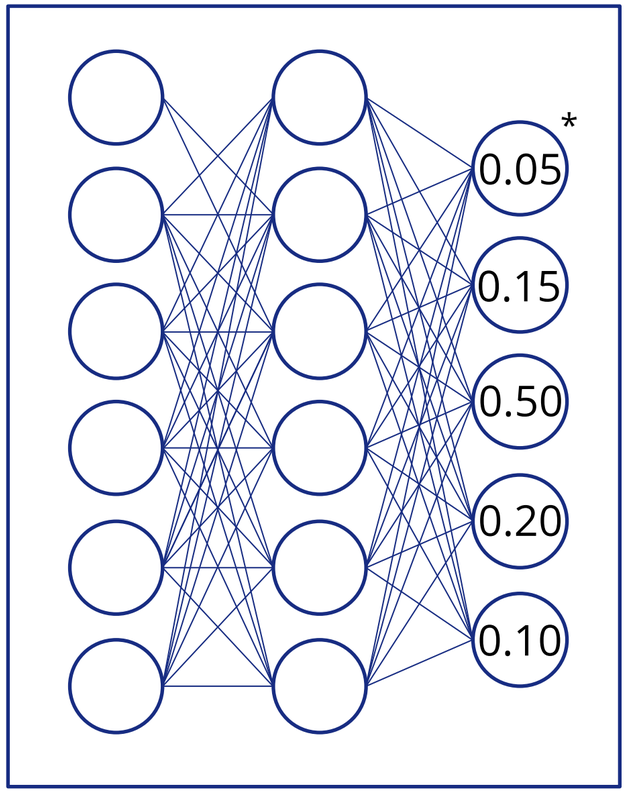
[2] Zhang et al.. "Generalized cross entropy loss for training deep neural networks with noisy labels." *Advances in neural information processing systems* 31 (2018).
* For simplicity, we omit the conversion from logits to probabilities

Baseline model

The standard loss function for multi-class categorical classification is the Cross-Entropy (CE) loss[2]



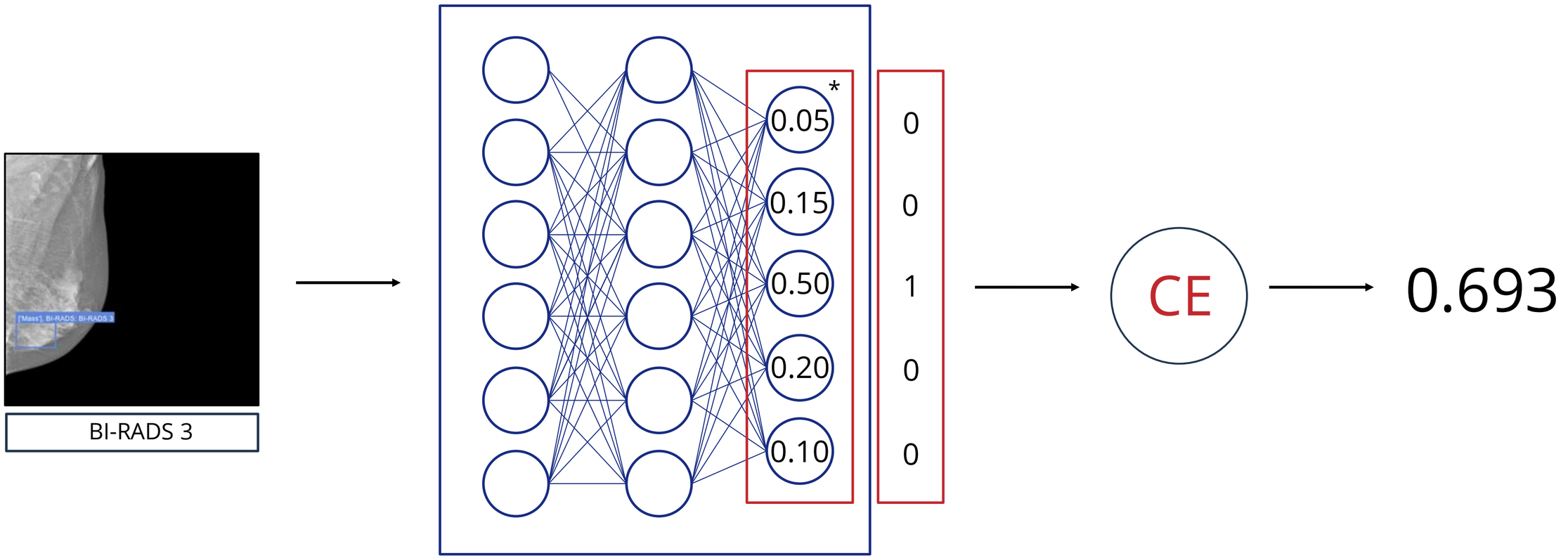
BI-RADS 3



[2] Zhang et al.. "Generalized cross entropy loss for training deep neural networks with noisy labels." *Advances in neural information processing systems* 31 (2018).
* For simplicity, we omit the conversion from logits to probabilities

Baseline model

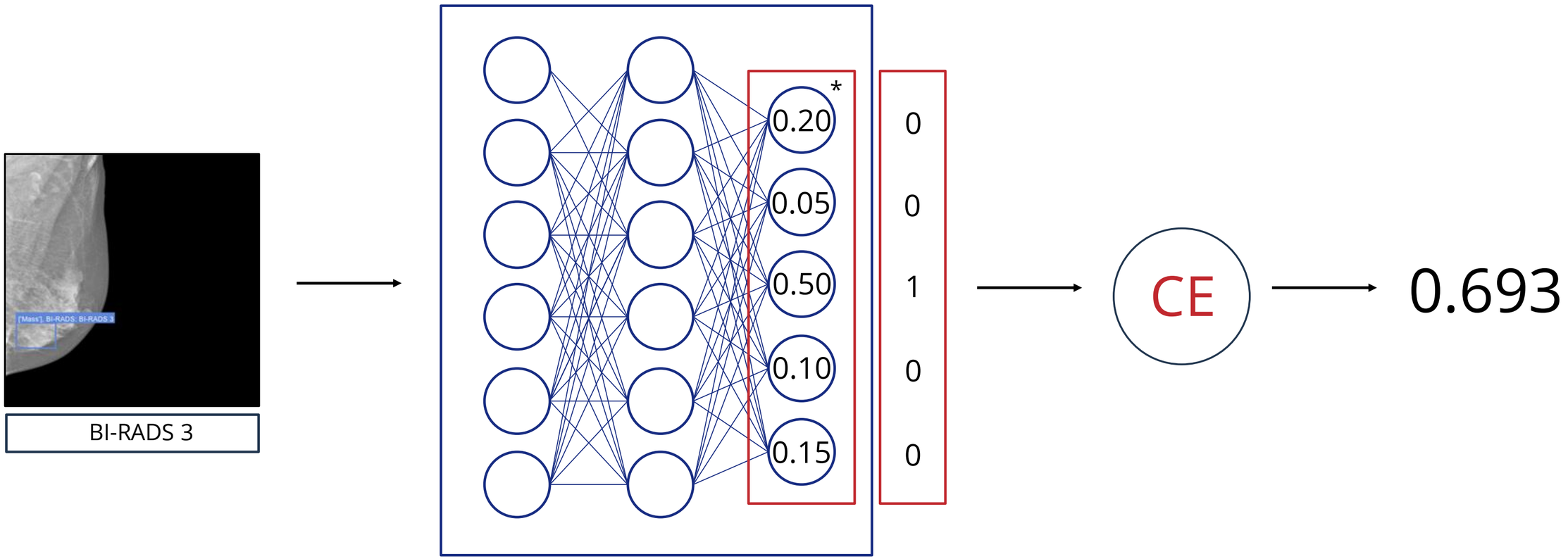
The standard loss function for multi-class categorical classification is the Cross-Entropy (CE) loss[2]



[2] Zhang et al.. "Generalized cross entropy loss for training deep neural networks with noisy labels." *Advances in neural information processing systems* 31 (2018).
* For simplicity, we omit the conversion from logits to probabilities

Baseline model

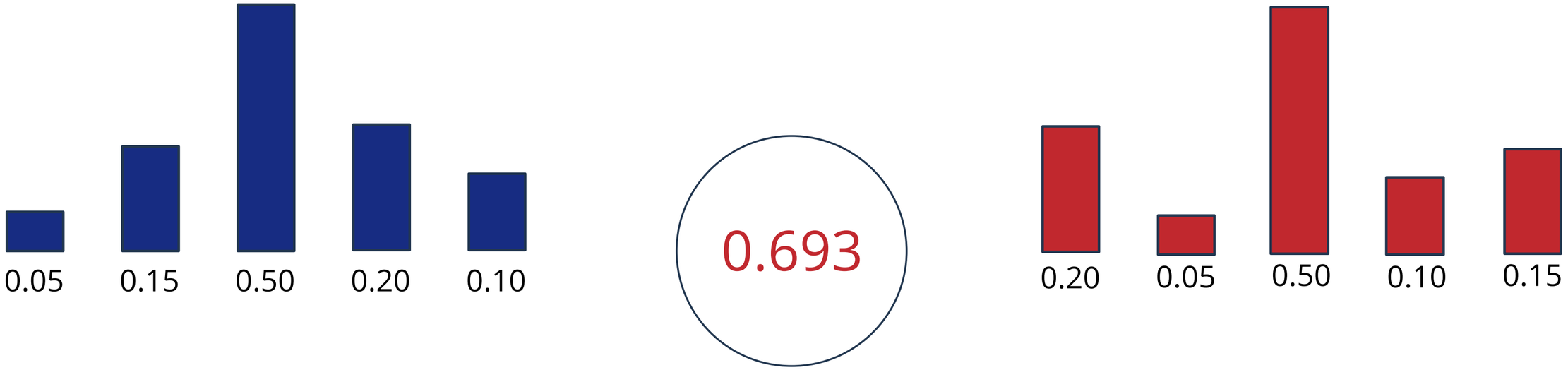
The standard loss function for multi-class categorical classification is the Cross-Entropy (CE) loss[2]



[2] Zhang et al.. "Generalized cross entropy loss for training deep neural networks with noisy labels." *Advances in neural information processing systems* 31 (2018).
* For simplicity, we omit the conversion from logits to probabilities

Baseline model

The standard loss function for multi-class categorical classification is the Cross-Entropy (CE) loss[2]



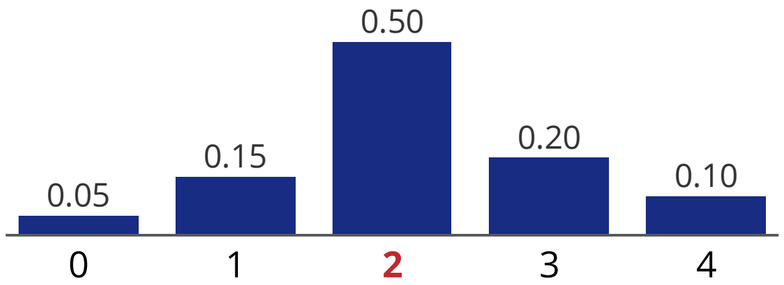
[2] Zhang et al.. "Generalized cross entropy loss for training deep neural networks with noisy labels." *Advances in neural information processing systems* 31 (2018).

Unimodality

A posterior is unimodal if the probabilities decrease monotonically moving away from the predicted class[1]

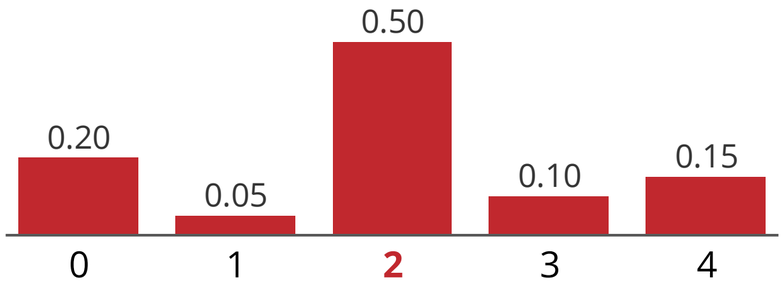
$$\hat{y}_{\hat{k}} \geq \hat{y}_{\hat{k}\pm 1} \geq \hat{y}_{\hat{k}\pm 2} \geq \cdots$$

Unimodal



CE = 0.693

Not unimodal



CE = 0.693

CE gives both posteriors the same loss: it depends only on the probability assigned to the true class

[1] Cardoso et al. "Unimodal distributions for ordinal regression." IEEE Transactions on Artificial Intelligence (2025).

Non-unimodal methods

The class order enters through the target encoding, not through the output shape

Nominal

Ignores the class order

CE

Non-unimodal

Uses the order; output shape unconstrained

OE

Soft-unimodal

Promotes unimodality through the training objective

Soft targets: BCE, Lp

CE + regulariser: Wu-Wass, MVL

Moment-based loss: exp_MSE

Hard-unimodal

Guarantees unimodality through the output layer

UN

Increasing constraint on the shape of the output distribution

[1] Cardoso et al. "Unimodal distributions for ordinal regression." IEEE Transactions on Artificial Intelligence (2025).

Ordinal models: Ordinal Encoding

Non-unimodal: the class order is used, output shape unconstrained

- Ordinal Encoding (OE) aims to learn the boundaries between the classes[3, 4]

[4] Cheng et al. "A neural network approach to ordinal regression." *2008 IEEE international joint conference on neural networks (IEEE world congress on computational intelligence)*. IEEE, 2008. [3] Frank & Hall. "A simple approach to ordinal classification." ECML (2001).

Ordinal models: Ordinal Encoding

Non-unimodal: the class order is used, output shape unconstrained

- Ordinal Encoding (OE) aims to learn the boundaries between the classes[3, 4]



[4] Cheng et al. "A neural network approach to ordinal regression." *2008 IEEE international joint conference on neural networks (IEEE world congress on computational intelligence)*. IEEE, 2008. [3] Frank & Hall. "A simple approach to ordinal classification." ECML (2001).

Ordinal models: Ordinal Encoding

Non-unimodal: the class order is used, output shape unconstrained

- Ordinal Encoding (OE) aims to learn the boundaries between the classes[3, 4]



BI-RADS 1: 0 0 0 0

BI-RADS 2: 1 0 0 0

BI-RADS 3: 1 1 0 0

BI-RADS 4: 1 1 1 0

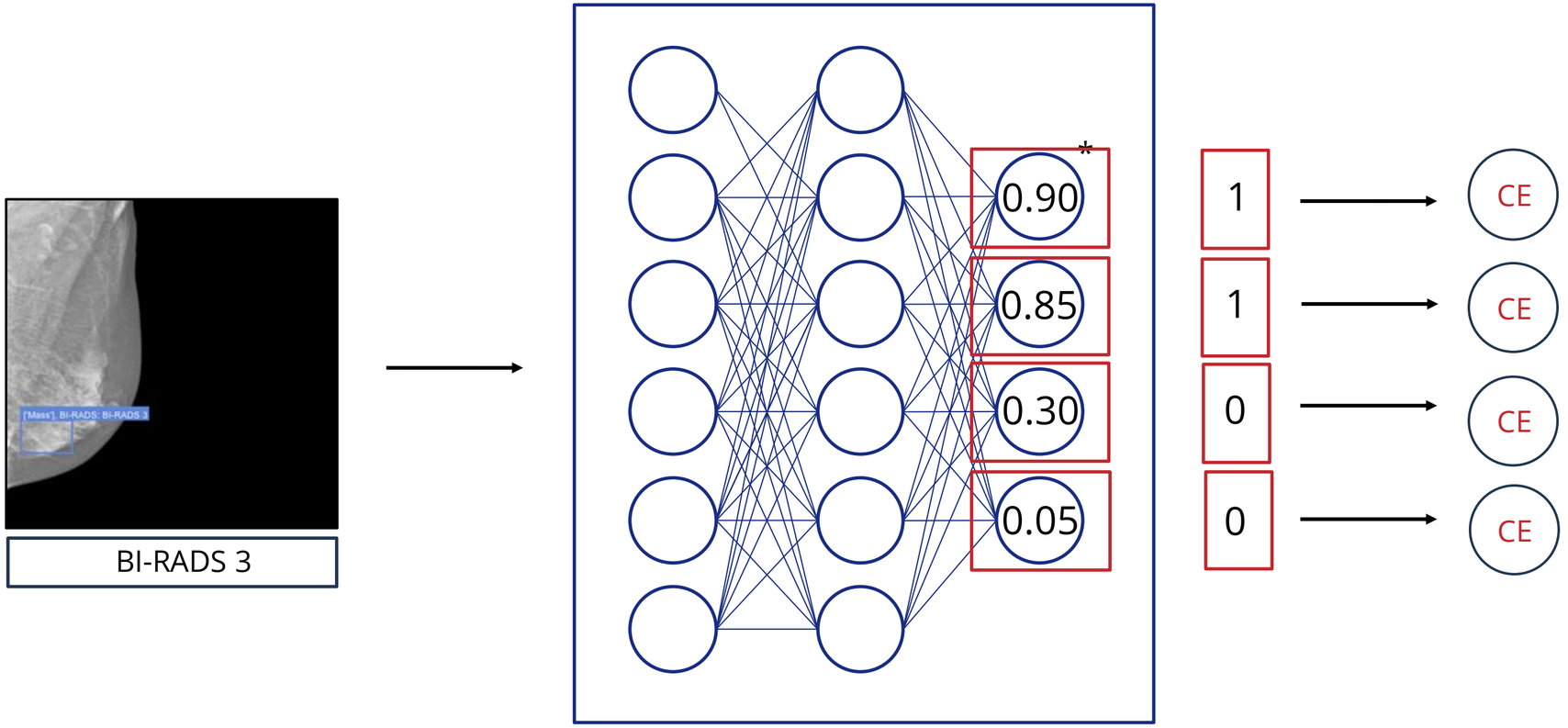
BI-RADS 5: 1 1 1 1

[4] Cheng et al. "A neural network approach to ordinal regression." *2008 IEEE international joint conference on neural networks (IEEE world congress on computational intelligence)*. IEEE, 2008. [3] Frank & Hall. "A simple approach to ordinal classification." ECML (2001).

Ordinal models: Ordinal Encoding

Non-unimodal: the class order is used, output shape unconstrained

- Ordinal Encoding (OE) aims to learn the boundaries between the classes[3, 4]



[4] Cheng et al. "A neural network approach to ordinal regression." *2008 IEEE international joint conference on neural networks (IEEE world congress on computational intelligence)*. IEEE, 2008. [3] Frank & Hall. "A simple approach to ordinal classification." ECML (2001).
* For simplicity, we omit the conversion from logits to probabilities

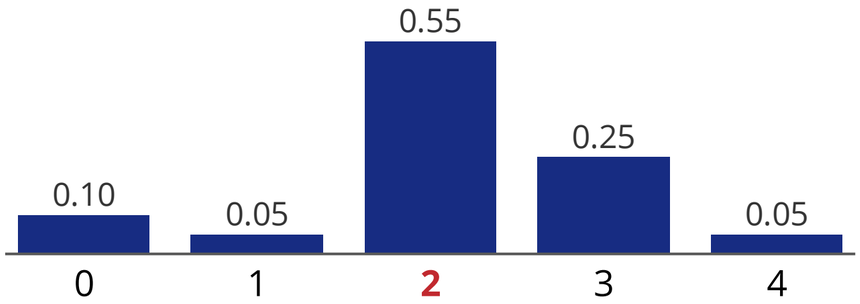
Ordinal models: Ordinal Encoding

At inference, class probabilities are reconstructed from the K − 1 binary outputs[3]

$$t_i = \mathbb{1}[\,i < k\,], \quad i = 0, \ldots, K-2 \qquad \sigma(g_i) \approx P(\,y > c_i\,)$$

$$q_0 = 1 - \sigma(g_0), \qquad q_i = \sigma(g_{i-1}) - \sigma(g_i), \qquad q_{K-1} = \sigma(g_{K-2})$$

σ(g) = [0.90, 0.85, 0.30, 0.05] → q



$q_0 > q_1$: the reconstructed posterior is not unimodal

- Each $\sigma(g_i)$ has its own binary loss; nothing forces them to decrease with i
- If $\sigma(g_1) < \sigma(g_2)$, then $q_2 < 0$

[3] Frank & Hall. "A simple approach to ordinal classification." ECML (2001).

Soft-unimodal methods

Unimodality is encouraged by the loss, but not guaranteed

Nominal

Ignores the class order

CE

Non-unimodal

Uses the order; output shape unconstrained

OE

Soft-unimodal

Promotes unimodality through the training objective

Soft targets: BCE, Lp

CE + regulariser: Wu-Wass, MVL

Moment-based loss: exp_MSE

Hard-unimodal

Guarantees unimodality through the output layer

UN

Increasing constraint on the shape of the output distribution

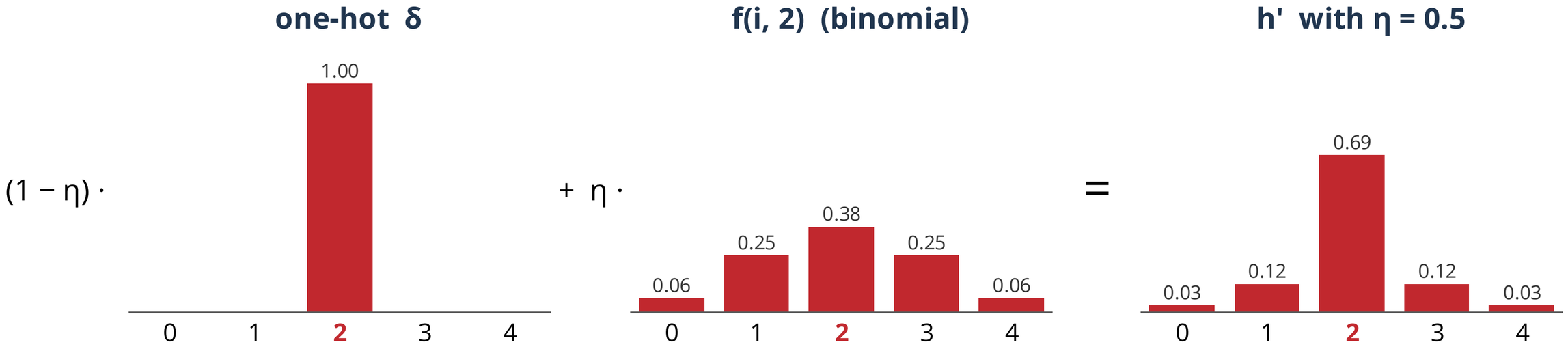
[1] Cardoso et al. "Unimodal distributions for ordinal regression." IEEE Transactions on Artificial Intelligence (2025).

Soft targets: label smoothing

Replace the one-hot target with a soft target centred on the true class, then train with CE[5]

$$h'(i,k) = (1-\eta)\,\delta_{i,k} + \eta\, f(i,k)$$

η ∈ [0, 1] sets the amount of smoothing; f(i, k) is a unimodal distribution with mode k



BCE and Lp differ only in the choice of f

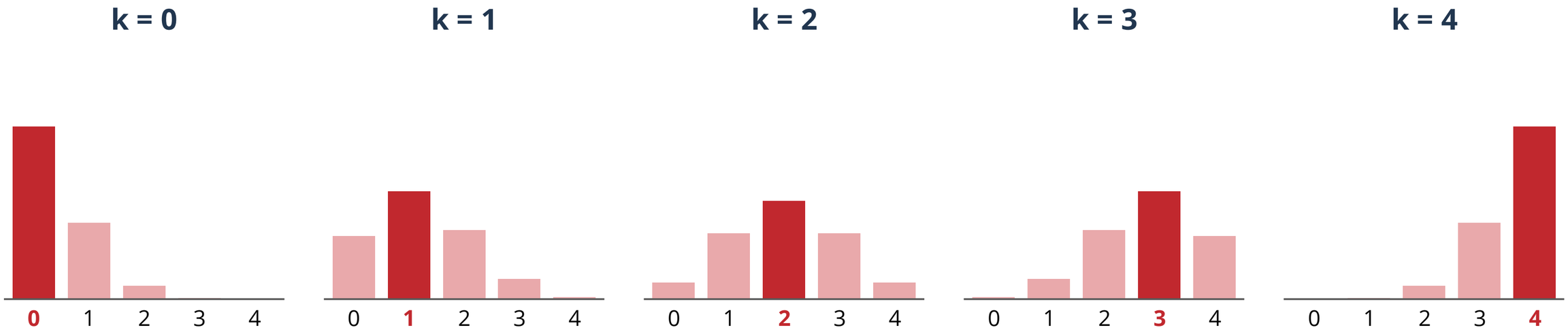
[5] Liu et al. "Unimodal regularized neuron stick-breaking for ordinal classification." Neurocomputing 388 (2020).

Soft targets: Binomial Cross-Entropy (BCE)

f(i, k) is a binomial distribution with K − 1 trials, with its mode at the true class[5, 6]

$$f(i,k) = \binom{K-1}{i} p_k^i \left(1-p_k\right)^{K-1-i}$$

p_k increases with k; in dlordinal, p_k is evenly spaced between 0.1 and 0.9



"BCE" here is Binomial Cross-Entropy, not binary cross-entropy. No hyperparameter besides η.

[5] Liu et al. "Unimodal regularized neuron stick-breaking for ordinal classification." Neurocomputing 388 (2020).

[6] Bérchez-Moreno et al. "dlordinal: A Python package for deep ordinal classification." Neurocomputing (2025).

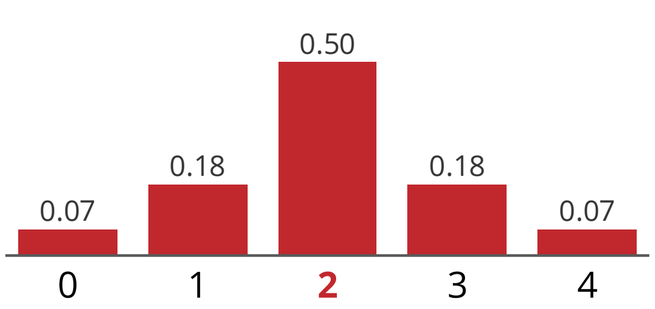
Soft targets: Lp (exponential) smoothing

f(i, k) decays exponentially with the distance to the true class[6, 7]

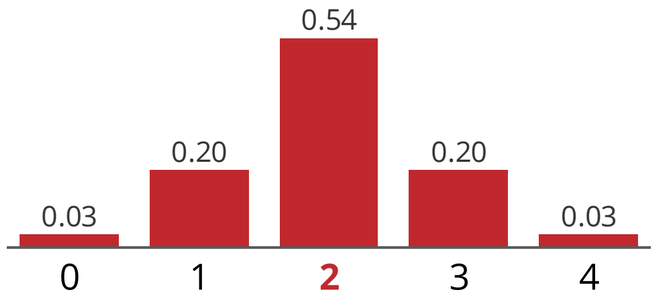
$$f(i,k) = \frac{\exp(-|i-k|^p)}{\sum_{j=0}^{K-1}\exp(-|j-k|^p)}, \qquad 1 \le p \le 2$$

Larger p shrinks the tails (|i − k| ≥ 2); p is a hyperparameter

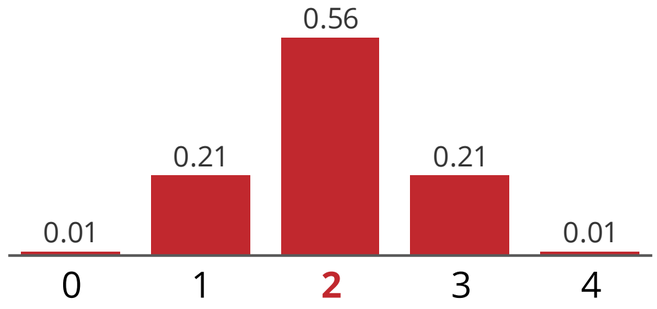
p = 1



p = 1.5



p = 2



Targets shown for k = 2

[6] Bérchez-Moreno et al. "dlordinal: A Python package for deep ordinal classification." Neurocomputing (2025). [7] Vargas et al. "Exponential loss regularisation for encouraging ordinal constraint to shotgun stocks quality assessment." Applied Soft Computing 138 (2023).

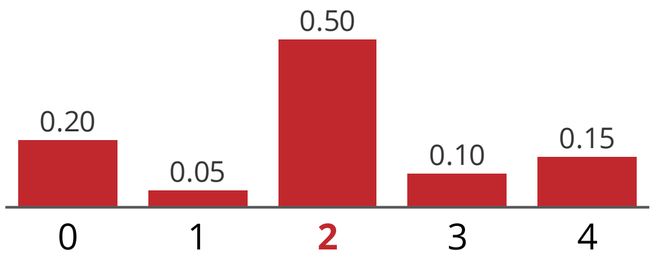
CE + regulariser: Wu–Wass

Penalise the distance between the prediction and its closest unimodal distribution with mode k[1]

$$\hat{\mathbf{y}}^P = \arg\min_{\mathbf{y}\in\mathcal{S}_k} D_1(\mathbf{y}, \hat{\mathbf{y}}) \qquad \mathcal{L}_{\text{Wu-Wass}} = \mathcal{L}_{\text{CE}} + \lambda\, D_2(\hat{\mathbf{y}}^P, \hat{\mathbf{y}})$$

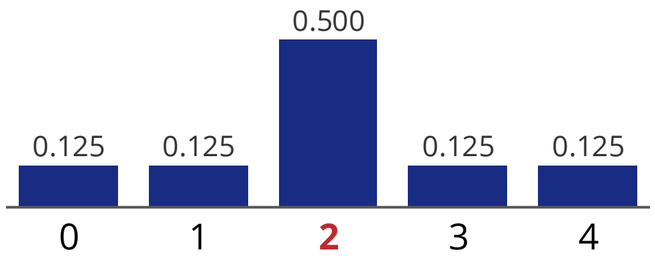
S_k: unimodal distributions with mode k; D_1: Wasserstein-1; D_2: Wasserstein-1 or KL divergence

ŷ (running example)



W_1 projection

ŷ^P



$D_1(\hat{y}^P, \hat{y}) = 0.10$

If ŷ is already unimodal with mode k, ŷ^P = ŷ and the penalty is 0

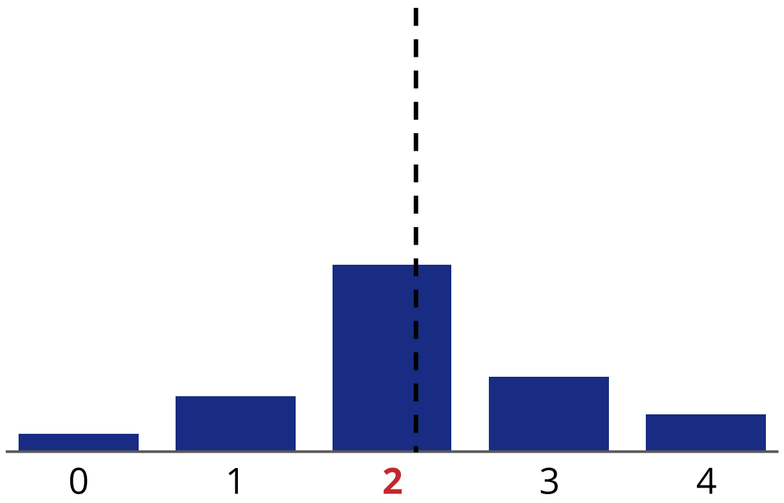
[1] Cardoso et al. "Unimodal distributions for ordinal regression." IEEE Transactions on Artificial Intelligence (2025).

Moments of the predicted distribution

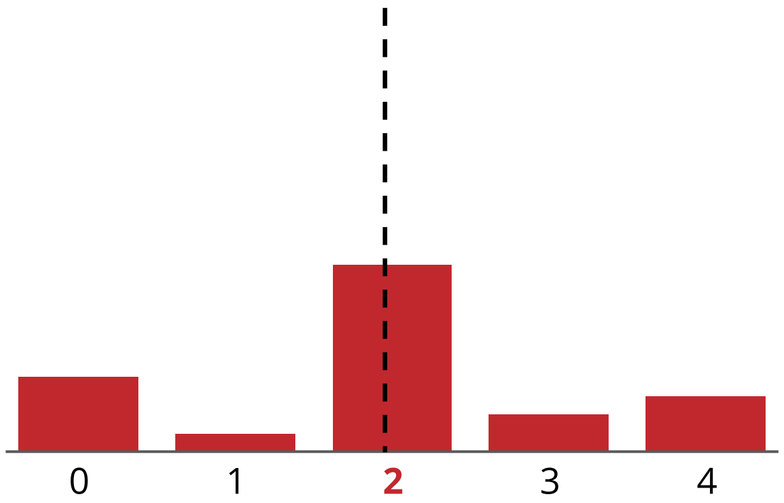
Treat the class index as a random variable distributed according to $\hat{y}$[8, 9]

$$\mu = \mathbb{E}[\hat{\mathbf{y}}|x] = \sum_{i=0}^{K-1} i\,\hat{y}_i \qquad \sigma^2 = \mathrm{Var}[\hat{\mathbf{y}}|x] = \sum_{i=0}^{K-1} \hat{y}_i\,(i-\mu)^2$$

μ = 2.15, σ² = 0.93



μ = 1.95, σ² = 1.55



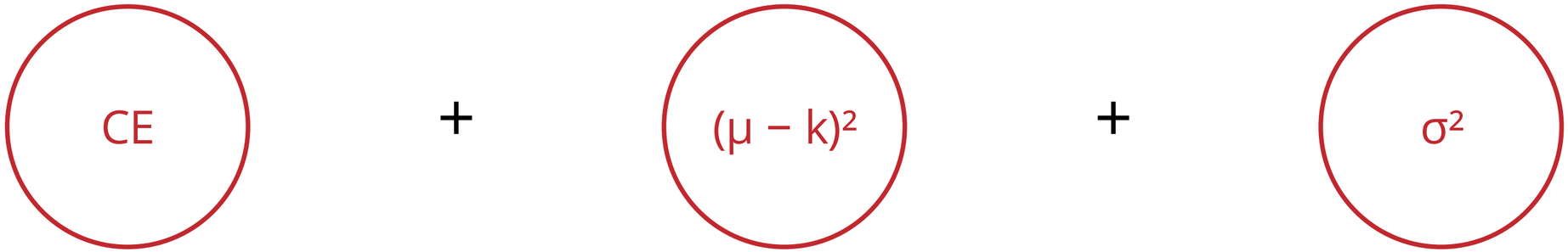
The mean alone places the non-unimodal ŷ closer to k = 2 (|μ − k| = 0.05 vs 0.15); the variance separates the two

[8] Pan et al. "Mean-variance loss for deep age estimation from a face." CVPR (2018). [9] Stelter, Corbetta et al. "Preserving ordinality in diabetic retinopathy grading through a distribution-based loss function." NLDL (2026).

CE + regulariser: Mean-Variance Loss (MVL)

Adds the two moments to CE as regularisation terms[8]

$$\mathcal{L}_{\mathrm{MVL}} = \mathcal{L}_{\mathrm{CE}} + \frac{\lambda_1}{2}\,(\mu - k)^2 + \lambda_2\,\sigma^2$$



fits the one-hot target

pulls the mean towards the true class

penalises spread around the mean

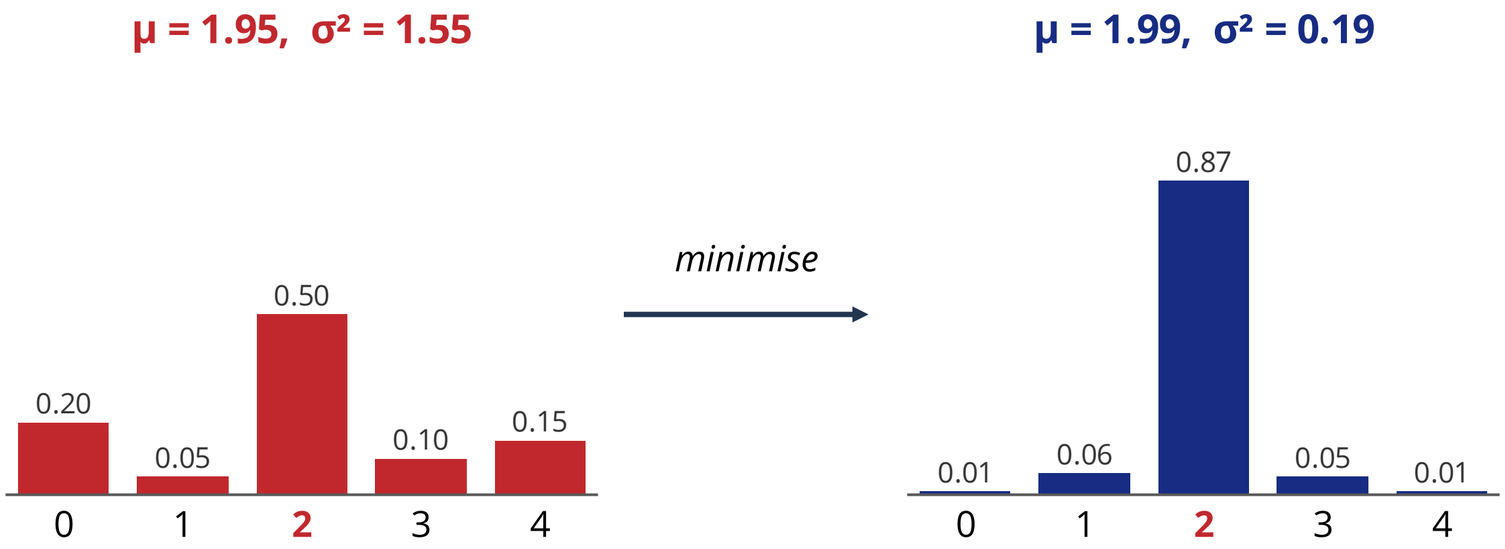
- Two hyperparameters (λ_1, λ_2) to tune jointly
- Originally proposed for age estimation from face images

[8] Pan et al. "Mean-variance loss for deep age estimation from a face." CVPR (2018).

Moment-based loss: exp_MSE

Optimises the two moments directly, without a CE term[9]

$$\mathcal{L}_{\text{exp_MSE}} = (\mu - k)^2 + \lambda\,\sigma^2$$



- One hyperparameter (λ)
- The loss is 0 only for the one-hot vector at k
- Running example: 0.0225 + 0.93λ (unimodal ŷ) vs 0.0025 + 1.55λ (non-unimodal ŷ)
- Edge classes (k = 0, K − 1) and inner classes are penalised differently[9]

μ → k and σ² → 0: the posterior approaches a one-hot vector at k

[9] Stelter, Corbetta et al. "Preserving ordinality in diabetic retinopathy grading through a distribution-based loss function." NLDL (2026).

Hard-unimodal methods

Unimodality holds for any input, by construction of the output layer

Nominal

Ignores the class order

CE

Non-unimodal

Uses the order; output shape unconstrained

OE

Soft-unimodal

Promotes unimodality through the training objective

Soft targets: BCE, Lp

CE + regulariser: Wu-Wass, MVL

Moment-based loss: exp_MSE

Hard-unimodal

Guarantees unimodality through the output layer

UN

Increasing constraint on the shape of the output distribution

[1] Cardoso et al. "Unimodal distributions for ordinal regression." IEEE Transactions on Artificial Intelligence (2025).

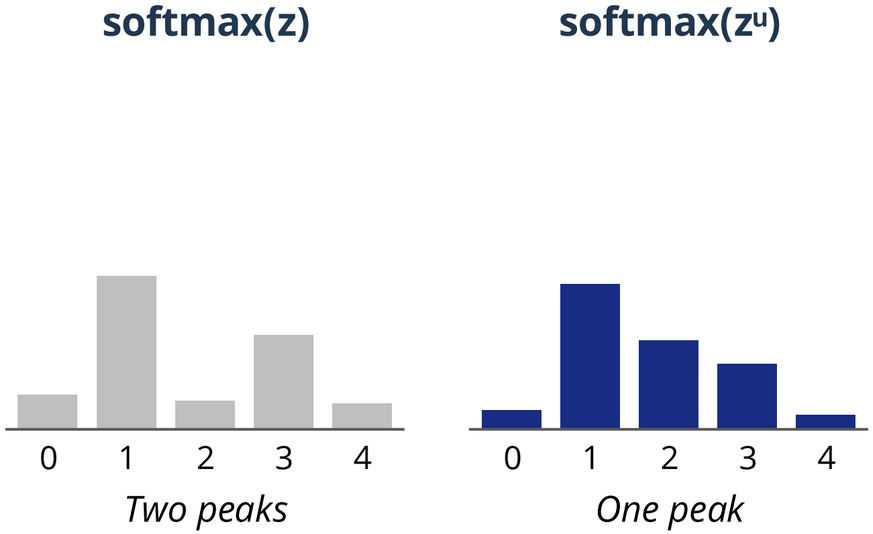
Hard-unimodal: UnimodalNet (UN)

The output layer makes the posterior unimodal for any logits[1]

$$z_i = f(v_i) \geq 0 \qquad z_i^{\ell r} = \sum_{j \leq i} z_j, \qquad z_i^{r\ell} = \sum_{j \geq i} z_j \qquad z_i^u = \min\left(z_i^{\ell r},\ z_i^{r\ell}\right), \qquad \hat{\mathbf{y}} = \mathrm{softmax}(\mathbf{z}^u)$$

f: a non-negative activation applied to the logits v

	i = 0	1	2	3	4
z	0.5	2.0	0.3	1.5	0.2
left-to-right sum	0.5	2.5	2.8	4.3	4.5
right-to-left sum	4.5	4.0	2.0	1.7	0.2
z^u = minimum	**0.5**	**2.5**	**2.0**	**1.7**	**0.2**



The minimum of a non-decreasing and a non-increasing sequence rises, then falls; softmax preserves the order

[1] Cardoso et al. "Unimodal distributions for ordinal regression." IEEE Transactions on Artificial Intelligence (2025).

Summary

Method	Family	What it changes	Hyperparameters
CE	Nominal	Baseline: one-hot target, softmax output	none
OE	Non-unimodal	Targets and output: $K - 1$ cumulative binary tasks	none
BCE	Soft-unimodal	Target: binomial soft labels	η
Lp	Soft-unimodal	Target: exponential soft labels	η, p
Wu-Wass	Soft-unimodal	Loss: CE + distance to closest unimodal distribution	λ
MVL	Soft-unimodal	Loss: CE + mean and variance terms	λ_1, λ_2
exp_MSE	Soft-unimodal	Loss: mean and variance terms only	λ
UN	Hard-unimodal	Output layer: cumulative sums and minimum	choice of f

Implementations and further reading

- **dlordinal**: Python package for deep ordinal classification, including binomial and exponential soft labels[6]
- **The methods presented are benchmarked in Stelter et al. Code for the methods in this tutorial**: github.com/Trustworthy-AI-UU-NKI/Ordinal-DR-Grading[9]
- **Taxonomy and unimodal methods** (including Wu-Wass and UN) in detail: Cardoso et al., IEEE TAI 2025[1]



Paper and code of Stelter et al!

[1] Cardoso et al., IEEE TAI (2025). [6] Bérchez-Moreno et al., Neurocomputing (2025). [9] Stelter et al., NLDL (2026).

Thank you

Questions?

References

[1] Cardoso et al. "Unimodal distributions for ordinal regression." IEEE Transactions on Artificial Intelligence (2025).

[2] Zhang & Sabuncu. "Generalized cross entropy loss for training deep neural networks with noisy labels." NeurIPS (2018).

[3] Frank & Hall. "A simple approach to ordinal classification." ECML (2001).

[4] Cheng et al. "A neural network approach to ordinal regression." IJCNN (2008).

[5] Liu et al. "Unimodal regularized neuron stick-breaking for ordinal classification." Neurocomputing 388 (2020).

[6] Bérchez-Moreno et al. "dlordinal: A Python package for deep ordinal classification." Neurocomputing (2025).

[7] Vargas et al. "Exponential loss regularisation for encouraging ordinal constraint to shotgun stocks quality assessment." Applied Soft Computing 138 (2023).

[8] Pan et al. "Mean-variance loss for deep age estimation from a face." CVPR (2018).

[9] Stelter, Corbetta et al. "Preserving ordinality in diabetic retinopathy grading through a distribution-based loss function." NLDL (2026).